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Copenhagen in Manhattan

In the winter of 1955, Werner Heisenberg gave a series of lectures at the University of St. Andrews in Scotland. The Cold War was in full swing; in the space of the previous decade, Heisenberg had gone from an enemy alien of the UK to a citizen of a trusted ally. Yet he was uneasy about his reputation among his fellow physicists and took the opportunity of his talk in Scotland to bolster it.

First, Heisenberg preached the familiar gospel of Copenhagen. “The idea of an objective real world whose smallest parts exist objectively in the same sense as stones or trees exist, independently of whether or not we observe them,” Heisenberg said, “is impossible.” How, then, does our world of stones and trees emerge from the world of atoms and molecules? “The transition from the ‘possible’ to the ‘actual’ takes place during the act of observation,” said Heisenberg. And what happens when we’re not looking? According to Heisenberg, that question can’t even be asked. “If we want to describe what happens in an atomic event, we have to realize that the word ‘happens’ can apply only to the observation, not to the state of affairs between two observations.” And what about the measurement problem? What makes observation so special? Whatever it is, it is “physical,” not “psychical,” said Heisenberg. “The transition from the ‘possible’ to the ‘actual’ takes place as soon as the interaction of the object with the measuring device, and thereby with the rest of the world, has come into play; it is not connected with the act of registration of the result by the mind of the observer.” Yet on the question of what constituted a “measuring